

Soumyadeep Nath

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EXPERIENCE

IIT Roorkee, Data Science Learner (Executive PG Programme) | Roorkee, India Feb 2026 – Present

- Engineered data pipelines and automated ETL workflows using Python (Pandas, NumPy) to clean, transform, and prepare structured and unstructured datasets for downstream analysis.
- Queried relational databases using advanced SQL — window functions, joins, subqueries, and aggregations — extracting actionable, analysis-ready business insights.
- Constructed Exploratory Data Analysis (EDA) frameworks applying statistical reasoning, hypothesis testing, and A/B testing methodology to validate data-driven decisions.
- Developed and evaluated supervised Machine Learning models — regression, classification, and tree-based ensembles (Random Forest, Gradient Boosting) — using scikit-learn.
- Implemented Generative AI applications by integrating LLM APIs, engineering structured prompts, and building RAG systems with embeddings and FAISS/Chroma vector databases.
- Translated complex model outputs into structured data narratives and interactive dashboards using Matplotlib, Seaborn, and Plotly; documented findings in reproducible Jupyter notebooks.

PROJECTS

AlignResume — AI-Powered Resume Optimization Platform May 2026

Live: [align-resume-beta.vercel.app](#) · GitHub: [github.com/sdn9300/align-resume](#) · Stack: Next.js · TypeScript · Tailwind CSS · Groq API (LLaMA 3.3 70B) · Playwright · Zod · Vercel

- Architected and deployed an end-to-end full-stack AI web application that parses resumes and job descriptions, scores alignment quality, and generates ATS-optimized bullet rewrites via a structured JSON prompt pipeline over the Groq API — with Zod schema validation ensuring deterministic, parseable LLM outputs.
- Implemented an automated truthfulness guardrail layer that validates rewritten bullets against the original resume, blocking export if hallucinated metrics, employers, or skills are detected — prioritizing honesty over inflation.
- Engineered Playwright-based headless PDF export, rate-limited API proxy (10 runs/hr per IP), and GitHub-Vercel CI/CD auto-deploy; deepened expertise in LLM application architecture, prompt engineering, and production-grade full-stack AI deployment.

SKILLS

Languages Python, SQL, TypeScript

Frameworks & Tools Next.js, Tailwind CSS, Shadcn UI, Streamlit, Vercel, Git / GitHub

Libraries NumPy, Pandas, Matplotlib, Seaborn, Plotly, scikit-learn, Zod

Machine Learning Regression, Classification, Random Forest, Gradient Boosting, Feature Engineering, Model Evaluation

Generative AI LLM APIs (Groq, OpenAI), Prompt Engineering, Structured JSON Pipelines, RAG, Embeddings, Vector Databases (FAISS / Chroma), Guardrail Design

Data Analytics EDA, Data Cleaning, Statistical Analysis, Hypothesis Testing, A/B Testing, Data Storytelling

Visualization Matplotlib, Seaborn, Plotly, Jupyter Notebook, Google Colab

Soft Skills Critical Thinking, Analytical Reasoning, Communication, Attention to Detail, Adaptability

CERTIFICATIONS

Machine Learning Statistical Foundations Professional Certificate LinkedIn Learning × Wolfram Research | Completed: Mar 2026 | Skills: Machine Learning · Linear Algebra · Wolfram Language

Anaconda Python for Data Science Professional Certificate LinkedIn Learning × Anaconda | Completed: Mar 2026 | Skills: Python · Data Analysis · Statistics

Career Essentials in Generative AI LinkedIn Learning × Microsoft | Completed: Mar 2026 | Skills: Generative AI · Microsoft Copilot · Responsible AI

EDUCATION

Indian Institute of Technology (IIT) Roorkee , Executive PG Programme in Data Science & AI – In Progress	Feb 2026 – Present
Presidency University, Kolkata , M.A. in English Literature Result: First Division	Oct 2022 – Jul 2024
Presidency University, Kolkata , B.A. Honours in English Literature Result: First Division	Sep 2019 – Jul 2022